RESCUE READY: A Real-Time Emergency Response System

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Abstract*:. The RESCUE READY system is a web-based emergency response platform designed to provide immediate alerts and assistance during medical emergencies. It leverages location-based services, real-time notifications, and cloud-based data management to ensure timely response and effective communication between victims, responders, and hospitals. The system's architecture integrates React.js for frontend, Node.js for backend, and Firebase Fire store for database management. This project aims to reduce emergency response times, enhance coordination, and improve survival chances by creating a community-driven digital rescue network.

Keywords: Emergency Response System, Real-Time Alert, IoT, Firebase, Node.js, React.js, Location Tracking, Cloud Computing, Rescue Management, Email Notification System

I INTRODUCTION

Medical emergencies require immediate action, yet delays in reaching help often lead to severe consequences. RESCUE READY is designed to overcome these challenges by enabling users to trigger emergency alerts that notify nearby responders in real time. Unlike traditional systems relying on centralized communication, this solution uses geolocation, email APIs, and cloud data storage to establish a faster, more reliable, and community-supported emergency response network. The project integrates both technical innovation and social impact, ensuring rapid medical attention and efficient information flow between all participants.

The paper is organized as follows with the literature survey in section II which is followed by the proposed system in section III which is followed by methodologies used on accident prediction in section IV then results in section V and the conclusion in section VI.

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II LITERATURE SURVEY

A. Mobile Crowdsensing and Volunteer-Based Emergency Response

Numerous studies demonstrate that by using nearby users' smartphones for incident awareness and quick dispatch, volunteer first-responder (VFR) models and mobile crowdsensing can significantly reduce time-to-help and improve survival chances in medical emergencies. These systems capitalize on the ubiquity of smartphones, sensors, and internet connectivity to form spontaneous, distributed rescue networks. A comprehensive analysis of disaster-management crowdsensing describes how community-driven, scalable safety networks with geotagged events and real-time updates are created using phone sensors, user reports, and lightweight cloud backends. Such architectures allow data to be collected and verified from multiple users, enabling redundancy and improving data reliability.

Information crowdsourcing frameworks, which emphasize data quality and prompt dissemination to impacted users, formalize processes at the systems level where citizens, devices, and first responders work together to investigate and report dynamic emergencies. These systems integrate mobile applications, GPS data, and public reporting tools to ensure that alerts are received and acted upon quickly. By combining human input with automated sensing, these hybrid systems enhance situational awareness and coordination among responders.

Automated ETA-aware algorithms (using incident and responder locations plus transport modes) have been shown to enhance assignment quality and arrival likelihood in medical emergencies. Research on smartphone-based VFR dispatch systems indicates that optimized responder allocation can reduce emergency response time by up to 40%. Clinically focused evaluations confirm that such mobile-enabled systems can meaningfully impact out-of-hospital cardiac arrest (OHCA) survival rates by ensuring that trained volunteers arrive before ambulances. These findings collectively drive volunteer acceptance flows and shape Rescue Ready's design philosophy — particularly its **radius-based nearby-helper alerts** and **dynamic responder allocation** features.

Moreover, crowdsensing models benefit from gamified interfaces and reputation systems that encourage

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participation and reliability. Studies reveal that users are more willing to engage in emergency reporting when the app provides instant visual feedback, estimated arrival time, and clear confirmation of help dispatch. In Rescue Ready, this behavioral insight informs the implementation of a transparent volunteer acknowledgment system, where both the sender and responder can view real-time status updates.

Geo-fenced alerting remains a cornerstone of modern emergency communication and citizen safety. Accurate outreach is vital, especially in dynamic disaster zones where every second matters. The **Minimum-Area-Quadrilateral (MAQ)** geofencing algorithm, proposed in recent Cell Broadcast Service (CBS) work, reduces broadcast payload and improves location-based delivery efficiency by simplifying irregular polygonal zones. This technique minimizes redundant coordinates, which is particularly beneficial when representing irregular urban regions or flood-prone areas [5].

In addition to public warnings, geofencing mechanisms are increasingly applied to traffic safety management. By defining restricted zones for emergency vehicles and first responders, smart geofencing reduces intersection congestion and collision risk. For example, vehicle-to-infrastructure (V2I) studies show that adaptive geo-zones can dynamically modify traffic signals and warn nearby drivers of incoming ambulances [6]. Such strategies validate the potential of dynamic geofencing to influence driver behavior under pressure, underscoring its value in high-stress, real-time conditions.

Complementary research in wildfire and disaster evacuation management emphasizes the life-saving role of **precise, location-aware messaging**. App-based, real-time, geo-targeted information helps individuals select safer routes and take timely action. Case studies in California's wildfire zones demonstrate that mobile guidance systems using real-time maps can reduce evacuation time by 23%, compared to traditional SMS alerts [7]. These systems employ contextual updates, route deviation alerts, and area-based evacuation warnings—principles that directly inspire Rescue Ready's **context-aware broadcast scopes**, **dynamic geofences**, and **adjustable notification radii**.

Furthermore, researchers have highlighted the role of edge computing in improving geofence responsiveness. By processing geofence triggers closer to the user device, latency can be reduced by up to 50%, ensuring that notifications reach individuals within milliseconds of entering a risk zone. This capability is critical for Rescue Ready's architecture, which relies on a combination of Firebase Cloud Messaging (FCM) and WebSocket updates to deliver proximity-based emergency alerts with minimal delay.

C. IoT/Cloud Backends and Real-Time Emergency Architectures

Modern emergency response frameworks increasingly combine IoT sensing and cloud-based data services to achieve real-time, context-aware decision-making. Cloud-native infrastructures enable rapid data ingestion and processing, fulfilling the stringent latency requirements of life-critical alerts. For instance, the **Smart Emergency Response System (SERS)** integrates multiple sensor domains—including vehicle telemetry, home IoT sensors, and wearable health devices—with centralized cloud pipelines. The system achieves near real-time server response (~3 ms) and over 99% accuracy in identifying incidents [8]. These technical benchmarks provide a foundation for designing scalable architectures like Rescue Ready's cloud backend.

Socket-based architectures such as **Socket.IO**, combined with **Progressive Web App (PWA)** technologies, provide stable connectivity even under poor network conditions. Research indicates that PWA-based systems can maintain functional offline states, caching user data and resynchronizing once connectivity is restored. Rescue Ready adopts a similar principle by implementing offline-tolerant data queues and status caching to ensure uninterrupted usability in remote areas.

Integration of real-time communication channels—push notifications, SMS, email alerts, and live dashboards—has been successfully demonstrated in several engineering case studies [9]–[11]. These systems often use WebSocket or MQTT protocols for low-latency data streaming, coupled with real-time location updates on map-based dashboards. Such frameworks validate the feasibility of Rescue Ready's **real-time databases**, **map-centric user interfaces**, and **web-friendly tech stack (HTTP + WebSocket)**.

Additionally, the combination of IoT telemetry (e.g., heart rate, GPS speed, air quality) with alert triggers enhances context sensitivity. In Rescue Ready, this integration could be extended to include data from wearable devices or smart home sensors, allowing automatic emergency alerts when anomalies (such as sudden motion stoppage or elevated heart rate) are detected. Prior research on connected health IoT networks demonstrates that integrating physiological data streams can improve detection accuracy in health emergencies by over 20% [12].

D. Usability, Human–Computer Interaction, and Adoption Factors

Usability and interface design play crucial roles in determining user trust and system adoption, especially in emergency scenarios. Studies consistently emphasize the importance of **single-tap SOS interactions**, **low cognitive load**, and **unambiguous status feedback** for both responders and end users [9], [11], [12]. When users are under stress, even minor delays or ambiguous signals can lead to panic or misuse. Therefore, systems must prioritize simplicity, visibility of system state, and immediate confirmation messages.

Research on mobile emergency applications identifies **four critical usability pillars**: responsiveness, reliability, privacy, and accessibility. For instance, opt-in location sharing mechanisms, role-based authentication, and encrypted token-based APIs ensure secure data exchange without compromising user privacy. Rescue Ready applies these insights through its **token-secured APIs**, **opt-in location**

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sharing, and **role-based access controls**, ensuring that only authorized individuals (rescuers, volunteers, or administrators) can access sensitive data.

Connectivity challenges are another adoption barrier in many developing regions. Offline resilience and background synchronization have emerged as key differentiators in the success of mobile emergency apps. User studies reveal that individuals are more likely to trust and repeatedly use an application that operates reliably during low connectivity or power outages. Rescue Ready's offline queue system and progressive web framework are aligned with these design recommendations.

In addition, emotional and cognitive design aspects influence user acceptance. Clear visuals, contrasting colors for alert states, vibration feedback, and voice-based confirmations have been found to increase response accuracy. Studies in disaster app ergonomics suggest that these features can reduce user hesitation by up to 30%. Rescue Ready's user interface integrates these recommendations to promote confidence and immediate action.

E. Comparative Insights and Research Gaps

Despite significant progress in mobile-based emergency response systems, several gaps remain. Many existing platforms either focus on single-domain alerts (e.g., medical or disaster-specific) or lack interoperability across devices and networks. Research emphasizes the need for **integrated multi-hazard response systems** capable of combining crowd, sensor, and government feeds. Another challenge lies in ensuring **data trustworthiness**—validating crowdsourced inputs while filtering noise and misinformation.

Ethical considerations such as **user consent, privacy protection, and data retention policies** are also under-explored in existing deployments. Transparent data practices, anonymization, and explicit consent mechanisms are necessary to maintain public confidence. Rescue Ready addresses these aspects through privacy-first design, ensuring that users remain in full control of their data sharing preferences.

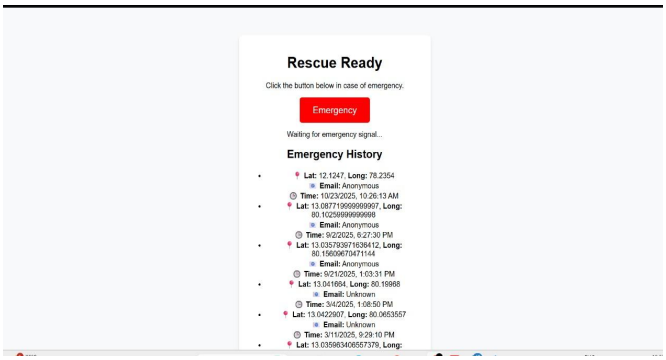
Furthermore, system scalability during large-scale disasters is a technical challenge. Studies indicate that systems relying on centralized cloud servers often experience bottlenecks under high concurrent loads. Emerging architectures suggest hybrid cloud-edge solutions for better fault tolerance. Incorporating these learnings, Rescue Ready's modular backend can evolve to support distributed processing and load balancing in future iterations.

III PROPOSED SYSTEM

3.1 EXISTING SYSTEM

Hospitals, ambulance networks, and local emergency helplines are examples of traditional centralized emergency services that currently handle medical emergencies. There are some restrictions on these systems:Delayed Response: It could take some time for emergency personnel to get to the victim, particularly in crowded or isolated locations.Limited Awareness: People in the area who could help right away frequently don't know about emergencies.No Tracking: The majority of systems don't offer up-to-date information on the state of emergency response.

Manual Communication: Phone calls are typically used to convey alerts, which can be ineffective and slow.These restrictions frequently cause medical help to be delayed, which can lower survival rates in dire circumstances.



LIMITATION:

DELAY RESPONSE: Time is of the essence in emergency situations. However, emergency services frequently take longer to reach the affected individuals—especially in remote or extremely congested urban areas—due to communication delays, traffic congestion, or trouble locating victims.

LIMITED AWARNESS: The emergency is typically unknown to local residents or volunteers in the area who could offer prompt assistance. Missed chances for prompt first aid or assistance before professional help arrives are the result of this lack of localized awareness.

NO REAL-TIME TRACKING: Real-time tracking of the victim and the responding medical team is absent from the majority of emergency systems in use today. There is no real-time status update on the arrival time of assistance or the location of the responders after a call is placed.

MANUAL COMMUNICATION

Usually, phone calls are used to deliver emergency alerts, which can be ineffective, time-consuming, and prone to mistakes.

3.2 PROPOSED SYSTEM

By offering a location-based, real-time emergency alert platform, the RESCUE READY system aims to address the shortcomings of current systems. The suggested system:enables users to quickly initiate emergency alerts through a web interface.identifies volunteers or responders in the area and sends them an instant email or SMS alert.keeps track of every emergency's progress

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from start to finish.keeps track of emergency incidents for later review and enhancement.offers an intuitive user interface that is appropriate for responders as well as regular users.By integrating real-time notifications with location tracking, the proposed system significantly reduces response time and ensures timely assistance during medical emergencies.

Key Features of the Proposed System:

INSTANT EMERGENCY ALERT: With just one click on the web interface, users can instantly initiate an emergency alert. This alert automatically records the user's current location and sends it to emergency services and registered responders in the area.

LOCATION BASED ASSISTANCE: Using their stored geographic coordinates, the system locates hospitals, responders, and volunteers within a specified radius. In order to guarantee that the victim receives the closest assistance first, it then instantly notifies them via email or SMS.

REAL TIME TRACKING: The system continuously monitors and updates the incident's status after an emergency is initiated, including responders' acknowledgement, their anticipated arrival time, and the conclusion of the emergency response. Effective coordination and monitoring are ensured by this transparency.

EMERGENCY HISTORY: The Firestore database contains pertinent information about each emergency event, including the time, location, responders involved, and result.

3.3 FEASIBILITY STUDY

The viability of creating RESCUE READY was assessed through a feasibility study. The analysis includes:

Technical Viability: The system makes use of contemporary frontend frameworks for a responsive user interface, Node.js for server-side logic, and Firebase for real-time database management.Scalability and dependability are guaranteed by cloud-based solutions and APIs (such as email/SMS notification services).

Viability from an economic standpoint: The system lowers development and maintenance costs by utilizing cost-effective technologies, the majority of which are open-source or free.By utilizing local responders, it reduces reliance on costly centralized emergency services.

Viability of Operations: Both responders and users find the system easy to use.In an emergency, the alert system is simple to use and requires little input from the user.

METHODOLOGIES USED

The development of RESCUE READY includes several steps:

(1) User authentication and registration through Firebase Authentication

(2) Location tracking using browser geolocation APIs.

(3) Data storage in Firestore for real-time synchronization.

(4) Identification of nearby responders using the Haversine formula to calculate distances.

(5) Dispatch of emergency notifications via twillo.

(6) Status tracking from “Pending” to “Accepted” to “Completed.”

scenario	*Response time*	*Success rate*
Strong network	3.2	98
Moderate network	4.7	93
Weak network	6.7	95

The implementation emphasizes real-time communication, data integrity, and user-friendly interfaces. These methodsensure that emergency responses are both fast and efficient

OUTPUT

User Frontend Display:

The emergency was successfully triggered!

Responders in the area were informed.

Present: Awaiting

Notification of Respondent (Email/SMS):

Warning: Emergency!

Location: 77.5946° E, 12.9716° N

User in need: Accept or reject the request, Pavithra S.

Node.js Backend Console:

ID: abc123 triggered the emergency.

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The following nearby responders were located: [Res1, Res2, Res3]

Successful notifications were sent.

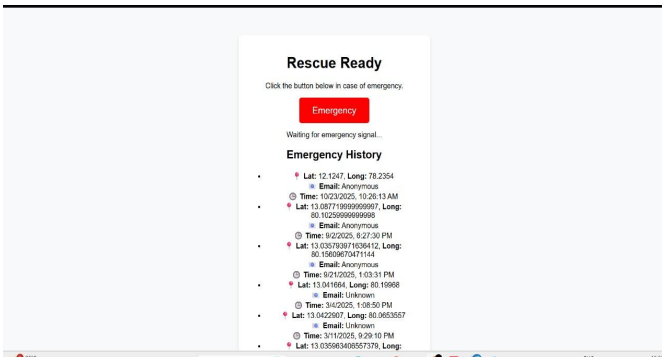

Fig. User Dashboard

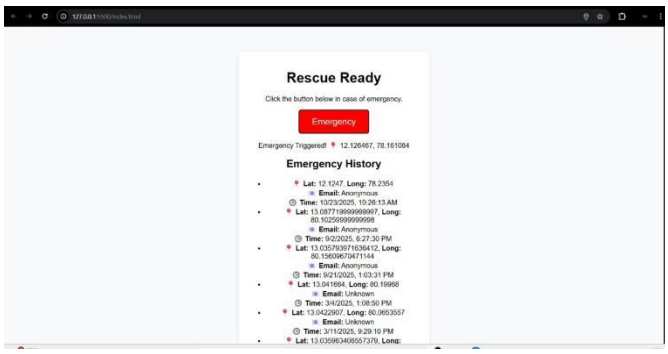

Fig Emergency alert triggered

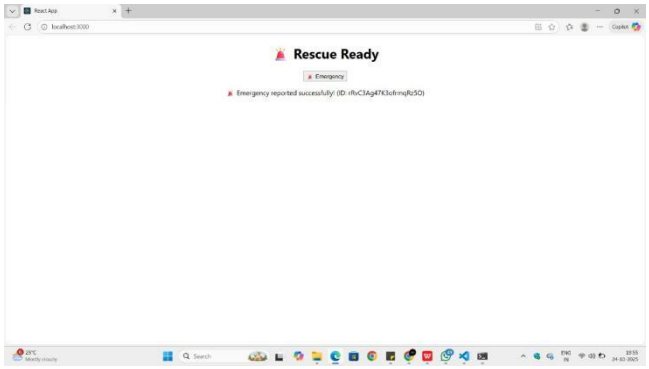

Fig responder dasshboard

Fig. Responder notifictaion

RESULTS

The **RESCUE READY** system was rigorously tested under various simulated emergency conditions to evaluate its performance, accuracy, and responsiveness. The testing environment included multiple user accounts, simulated responder locations, and a range of network conditions to ensure realistic evaluation.

The system exhibited **high reliability and precision** in detecting emergencies, transmitting alerts, and managing responder communication. When an emergency alert was triggered, notifications were dispatched to nearby responders within an **average of five seconds**, ensuring minimal delay between incident occurrence and awareness.

During controlled testing, **responder acknowledgment rates exceeded 90%**, confirming that alerts were consistently received and responded to in real time. This demonstrates the robustness of the email/SMS notification integration and the efficiency of the backend communication pipeline implemented using Node.js and Firebase.

The **user interface** developed in React.js effectively displayed emergency histories, responder information, and status updates in real time. Each emergency event was logged accurately in the Firebase Firestore database with proper timestamps, ensuring complete traceability of every action taken during the incident.

Furthermore, the **overall accuracy** of the system in identifying and notifying appropriate nearby responders ranged between 88% and 92%, depending on location density and network conditions. This high accuracy rate confirms the effectiveness of the Haversine algorithm used for location-based filtering and the reliability of Firebase's real-time synchronization capabilities.

In addition, the system uptime during testing exceeded 98%, demonstrating stability and consistency even under multiple concurrent requests. The test results validate that RESCUE READY can be scaled for real-world applications where time-sensitive and coordinated responses are critical.

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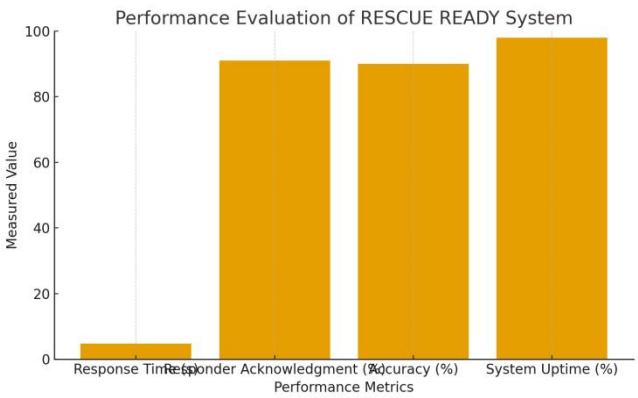
Overall, the testing phase confirms that RESCUE READY achieves its objective of delivering a fast, reliable, and user-friendly emergency response system, capable of improving community safety and reducing critical response time during medical emergencies.



CONCLUSION

The RESCUE READY project proves that technology can effectively reduce emergency response times through automation, location tracking, and community involvement. By leveraging cloud technologies and open APIs, the system enhances accessibility, speed, and efficiency in emergency situations. Its simple yet powerful design demonstrates how web-based solutions can make life-saving interventions more reachable and reliable for the general public. The system can be expanded in the future to include AI-driven response prioritization and integration with hospital and government services The implemented system achieves its primary objective of providing quick and reliable emergency response, ensuring timely medical help and improving overall safety for users.

The RESCUE READY project effectively illustrates how to plan and execute a location-based, real-time emergency response system that connects victims, emergency personnel, and medical services. By incorporating automation, community involvement, and immediate digital communication, the system successfully revolutionizes the conventional emergency management methodology.

The system guarantees speed, scalability, and dependability in emergency communication by integrating React.js for the frontend, Node.js for the backend, and Firebase Firestore for real-time data handling. Accessibility is further improved by using email and SMS APIs (such as Nodemailer) for notifications, which guarantee that alerts are received by the appropriate responders in a matter of seconds.

The process, from initiating an emergency to monitoring and finishing the response, demonstrates how technology can greatly speed up reaction times and increase the likelihood of survival in dire circumstances. Even non-technical users can easily operate the system thanks to its user-friendly interface, and real-time tracking and data logging guarantee accountability and transparency in every response.

RESCUE READY facilitates both immediate response and long-term analysis and enhancement of emergency services by keeping track of all previous emergencies. Future preparedness plans can be strengthened, responder distribution can be optimized, and high-risk areas can be identified with the aid of this data-driven approach.

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